AAKASH E

+91-8148087518 · Hosur, India · eaakash8805@gmail.com
linkedin.com/in/aakashe08 · github.com/AakashE08

Professional Summary

Computer Science & Engineering undergraduate (Specialization in AI & ML) specializing in robotics, autonomous UAV navigation, and aerial systems. Hands-on experience at IIT Hyderabad's TiHAN hub engineering flight control software and state estimation pipelines for GPS-denied environments, backed by a strong foundation in computer vision and SAR spatial analytics.

Education

SRM Institute of Science and Technology, Kattankulathur, TN — 2023 – 2027
B.Tech in Computer Science and Engineering (Specialization in AI & ML) — **CGPA: 8.65/10**

Skills

Robotics / UAV Systems: ROS, Gazebo, RViz, MAVROS, MAVLink, FAST-LIO, Optical Flow
Machine Learning / CV: TensorFlow, PyTorch, scikit-learn, OpenCV, CNNs, Transformers, YOLOv8
Programming Languages: C++, Python, Java
Cloud & DevOps: AWS, Docker, Kubernetes, Git
GitHub
Databases & Storage: MongoDB, MySQL, DynamoDB

Experience

Intern – Autonomous Navigation of UAVs — May 2026 – Aug 2026
NMICPS Technology Innovation Hub on Autonomous Navigation Foundation (TiHAN), IIT Hyderabad

- Completed a three-month internship (20 May – 19 Aug 2026) at TiHAN, IIT Hyderabad's autonomous navigation technology hub, in the area of autonomous navigation of unmanned aerial vehicles (UAVs).
- Worked on a project involving autonomous drone navigation in GPS-denied indoor environments, contributing to system development, integration and testing.
- Gained hands-on experience with UAV software stacks, simulation and visualization tools, and onboard-to-ground-station communication.

Projects

Oil Spill Detection using Sentinel-1 SAR with Tiling and Anomaly Detection

- Designed a scalable SAR-based oil spill detection framework using Sentinel-1 GRD imagery (VV, VH).
- Developed a tiling-based anomaly detection pipeline with localized statistical modeling to address scalability in full-scene SAR processing.
- Implemented tile-wise intensity distribution modeling to adapt to spatially varying backscatter conditions.
- Built a two-stage detection system combining anomaly scoring with GLCM-based region-level refinement and spatial filtering.
- Generated GIS-ready outputs for large-scale marine monitoring and spatial analysis.

Edge-Optimized Real-Time Object Detection (YOLOv8)

- Developed a real-time object detection system optimized for edge deployment on NVIDIA Jetson Nano.
- Integrated TensorRT and ONNX pipelines to enable low-latency, GPU-accelerated inference under resource constraints.
- Designed a live video processing pipeline with bounding box visualization, confidence scoring, and FPS monitoring.

CNN Feature Map & Activation Analysis Tool

- Developed an interactive tool for visualizing CNN feature maps and intermediate layer activations.
- Implemented forward-hook mechanisms to extract and analyze internal network representations.
- Enabled model interpretability through heatmap-based visualization of learned features.

Certifications

MongoDB Associate Developer — Credly Verification
SAP Certified – SAP Generative AI Developer — Credly Verification
SAP Certified – Data Analyst (SAP Analytics Cloud) — Credly Verification
Google Cloud Engineering Certificate — Credly Verification